

Spectral Partitioning as a Deterministic Redistricting Construction Baseline

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Draft — May 12, 2026

Abstract

This draft studies spectral graph partitioning as a deterministic construction baseline for congressional redistricting. The paper focuses on the staged T.14 implementation: Fiedler-vector approximation, population-balanced sweep cuts, recursive proportional target handling, and RPLAN audit sidecars through the BISECT CLI. The central claim is modest: spectral partitioning is a transparent deterministic baseline for comparison. RPLAN/RCTX sidecars verify declared profile and lineage, not fairness, legal compliance, exact optimality, or a replacement for METIS-family construction.

1 Introduction

Spectral graph partitioning is a natural deterministic baseline for redistricting because it exposes a continuous relaxation of the cut problem. This paper positions T.14 as a transparent construction method whose value is explainability, reproducibility, and comparison against stronger construction and optimization families.

2 Method

The staged method approximates a Fiedler vector, sorts units by the spectral coordinate, and evaluates population-balanced sweep cuts. Recursive calls pass proportional target fractions so odd and non-power-of-two district counts are handled without silently changing the intended seat allocation.

In the atlas framing, the candidates are not arbitrary graph cuts. They are prefix cuts induced by the Fiedler ordering. The ordering explains why the candidate set exists; the sweep table explains why a particular prefix is selected or skipped. The downstream consequence of the selected root cut is a pair of recursive workloads with explicit left/right seat counts.

2.1 Reproducible Procedure

The constructor operates on the tract-adjacency graph supplied by the BISECT state pipeline. Vertices carry population weights, edges encode adjacency, and the current implementation

treats edge weights as the graph weights available from that pipeline. For each recursive subproblem:

1. Build the weighted graph induced by the current unit set.
2. Form the Laplacian used by `bisect-apportion::spectral`.
3. Approximate a Fiedler vector and sort units by spectral coordinate, breaking ties by stable unit identifier.
4. Sweep prefixes in that order and select the cut whose population is closest to the requested proportional target while satisfying the declared tolerance.
5. Recurse on the two induced subgraphs with the requested left and right seat counts until each leaf represents one district.

This procedure makes determinism an implementation contract: the same graph, weights, target fractions, and tie-breaking identifiers must produce the same assignment and lineage. The graph assumptions are deliberately narrow. Water gaps, islands, disconnected components, and alternative adjacency definitions belong to the input graph construction layer, and near-symmetric graphs can make the spectral coordinate weakly informative even when the run remains deterministic.

3 Implementation

The implementation lives in `bisect-apportion::spectral` and is exposed through `bisect state --structure spectral` and YAML `structure: spectral`. Final plan paths use the shared RPLAN/RCTX audit sidecar flow when emitted by the CLI.

4 Audit Artifacts

When spectral construction is exported as a final plan, the paper follows the shared RPLAN/RCTX fixed point: assignments, algorithm lineage, parameter hashes, and validation results are written as sidecars that can be checked by the verifier without re-running the constructor. The audit claim is limited to the declared profile and does not imply political fairness or legal compliance. The audit package also does not prove that the spectral cut is unique, globally optimal, or preferable to another valid construction family.

5 Evaluation Plan

The first evidence layer is fixture correctness: path graphs, two-clique graphs, determinism, and recursive split handling. The next evidence layer should compare spectral construction against METIS, GeoSection, and clustering on selected real-data smoke cases.

Claim	Evidence now	
Deterministic fixture output	L0 path/two-clique tests	I
Odd- k recursion handling	Recursive split fixtures	I
RPLAN audit lineage	L1 sidecar checks, method package, and 100-unit benchmark smoke	I
Quality vs. METIS/GeoSection	None in this paper	S
Runtime scaling	None in this paper	I

Table 1: Evidence ladder for T.14 claims. Fixture correctness, audit checks, and empirical plan quality are separate evidence levels.

Constructor	Primary role	Exposed failure mode	Audit surface
Spectral	deterministic baseline	weak cut or imbalance	lineage and validation
METIS	multilevel baseline	dependency and heuristic variance	manifest lineage
GeoSection	geographic baseline	geometric oversimplification	assignment checks
Capacity clustering	capacity baseline	infeasible capacity/repair	status sidecars
Regionalization	merge baseline	merge-order sensitivity	merge witnesses

Table 2: Planned constructor comparison framing. Quantitative entries remain future work until commands, data, and outputs are recorded.

6 Limitations

Spectral partitioning is a relaxation. A valid and deterministic spectral plan does not imply global optimality, political fairness, or legal compliance beyond the declared audit profile.

The method is also sensitive to graph construction choices. Disconnected components, weakly connected geography, nearly symmetric graphs, and edge weight conventions can change the spectral coordinate and therefore the sweep order. Those are inspectable inputs to the audit trail, but they are not solved by determinism alone.

7 Conclusion

T.14 adds a cheap, deterministic construction baseline to the BISECT family. Its publication value is clearest when it is presented as an auditable point of comparison rather than as the strongest available optimizer. The fixed-point audit contract makes the output reproducible under a declared profile, while the evidence ladder keeps fixture correctness, sidecar validation, and future empirical quality claims separate.

References

BISECT Project. T.14 spectral partitioning implementation spec, 2026. BISECT internal specification.